Unbounded two-cycle contributions to permanents of positive semidefinite matrices

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Abstract. Let A be a complex Hermitian positive semidefinite matrix of order n with a distinguished index 0, and let $T(A) = \sum_{j\neq 0} |a_{0j}|^2 \operatorname{per} A\big(\hat{0},\hat{j}\big)$ be the contribution to $\operatorname{per} A$ of the permutations in which 0 lies in a two-cycle. Zhang's 2016 survey records the inequality $T(A) \le \operatorname{per} A$ as an open problem. We show that it fails, and that it fails by an unbounded factor: $T(A)/\operatorname{per} A$ is unbounded over complex matrices of rank two and over real matrices of rank three, and an explicit sequence of complex rank-two matrices of order n has $T(A)/\operatorname{per} A \sim (c_0/3)\log n$ with $c_0 = 0.4253\ldots$. A 13×13 Gaussian-integer Gram matrix of rank two with $T(A) > \operatorname{per} A$ is exhibited, as is a real integer matrix of order 473 and rank three. The proof passes through the Fischer norm on polynomials: $\operatorname{per} A$ is the squared norm of a product of linear forms, $T(A)$ is a weighted sum of squared norms of the product with one factor removed, and a univariate family built from the sine product has the required logarithmic growth.

1. Introduction

Let $A = (a_{ij})$ be a complex Hermitian positive semidefinite matrix of order n, indexed by $0,\ldots,n-1$, and let $\operatorname{per} A = \sum_{\sigma\in S_n}\prod_i a_{i,\sigma(i)}$. For distinct indices i, j write $A(\hat{\imath},\hat{\jmath})$ for the principal submatrix obtained by deleting rows and columns i and j, and $A(\hat{\imath})$ for the deletion of i alone; the permanent of the empty matrix is 1. Group the permutations according to the cycle containing 0. Those fixing 0 contribute $a_{00}\operatorname{per} A\big(\hat{0}\big)$; those in which 0 lies in a two-cycle with j contribute $a_{0j}a_{j0}\operatorname{per} A\big(\hat{0},\hat{j}\big) = |a_{0j}|^2 \operatorname{per} A\big(\hat{0},\hat{j}\big)$; the remaining permutations place 0 in a cycle of length at least three. Put

$$T(A) = \sum_{j=1}^{n-1} |a_{0j}|^2 \operatorname{per} A\big(\hat{0},\hat{j}\big), \qquad R(A) = \frac{T(A)}{\operatorname{per} A}. \tag{1}$$

Thus $T(A)$ is the two-cycle contribution to the permanent, and $\operatorname{per} A - a_{00}\operatorname{per} A\big(\hat{0}\big) - T(A)$ is the contribution of the long cycles.

The fixed-point contribution never exceeds the permanent: $a_{00}\operatorname{per} A\big(\hat{0}\big) \le \operatorname{per} A$ is the one-by-one case of Lieb's inequality [1]. Every term of $T(A)$ is nonnegative. The long-cycle terms carry no sign, so the corresponding question for the two-cycle contribution,

$$T(A) \le \operatorname{per} A, \tag{2}$$

is not settled by positivity. Drury proposed in 2016 the stronger inequality $\big(a_{00}\operatorname{per} A\big(\hat{0}\big)\big)^2 + T(A)^2 \le (\operatorname{per} A)^2$, which implies Chollet's conjecture; Hutchinson [2] refuted it with a 4×4 complex correlation matrix. The weaker inequality (2) survived. Zhang's survey [3] states it as equation (21) and records: "(21) itself is an interesting open problem."

This paper shows that (2) fails, and that no constant can be put in its place.

Theorem 1.1.

1. *$R(A)$ is unbounded over complex Hermitian positive semidefinite matrices of rank two. Explicitly, the matrices A_L of Section 4, of order $n_L = 64L^3 + 4L + 1$ and rank two, satisfy*

$$R(A_L) \sim \left(\frac{c_0}{3}\right) \log n_L \quad (L \to \infty), \quad c_0 = \frac{4}{9}\frac{\|f_\infty\|^2}{\|zf_\infty\|^2} = 0.4253..., \tag{3}$$

   where $f_\infty(z) = \sin(\pi z^2/9)/(\pi z^2/9)$ and the norms are Fischer norms.
2. *$R(A)$ is unbounded over real symmetric positive semidefinite matrices of rank three, and over such matrices with integer entries.*
3. *In both cases the matrices can be taken to be correlation matrices, and the unboundedness persists over positive definite matrices.*

The failure of (2) is already visible in a single small matrix.

Theorem 1.2. *Let A be the Gram matrix of the thirteen vectors*

$$(1,0); \quad (5, 3i^j) \quad (0 \le j \le 3); \quad (5, 5(1+i)i^j) \quad (0 \le j \le 3); \quad (0,1) \text{ four times}$$

in $\mathbb{C}^2$, with $(1,0)$ distinguished. Then A is Hermitian positive semidefinite of rank two with Gaussian-integer entries, and

$$\operatorname{per} A = 32\,030\,415\,900\,000\,000\,000, \quad T(A) = 32\,217\,583\,221\,000\,000\,000, \tag{4}$$

so that $R(A) = 1193243823/1186311700 > 1$.

A real symmetric integer matrix of order 473 and rank three with $R > 1$ is given in Section 7.

Method. Write A as the Gram matrix of vectors $v_0, \dots, v_{n-1}$ and let ℓ_i be the linear form with coefficient vector v_i. In the Fischer inner product on polynomials, $\operatorname{per} A = \|\ell_0 p\|^2$ with $p = \prod_{i \ge 1} \ell_i$, and $\operatorname{per} A\big(\hat{0}, \hat{\jmath}\big) = \|p/\ell_j\|^2$. So $R(A)$ compares the norm of a product of linear forms with the norms of the products obtained by deleting one factor, weighted by $\big|\langle v_0, v_j\rangle\big|^2$. For a univariate polynomial f with roots α_j the analogous ratio is

$$Q(f) = \frac{\sum_j \|f/(z-\alpha_j)\|^2}{\|zf\|^2}. \tag{5}$$

For $f_L(z) = \prod_{k=1}^{L}(1 - z^4/81k^2)$, whose roots are $3\sqrt{k}\, i^j$, each deleted-root norm is $\|f_L\|^2/|\alpha|^2$ up to a summable error, and summing $|\alpha|^{-2} = 1/9k$ over the roots produces the harmonic number: $Q(f_L) \sim c_0 \log L$. Two transfers convert $Q(f)$ into $R(A)$: a homogenization with many copies of one vector gives complex Gram matrices of rank two, and a Gaussian concentration argument on the polynomial $xy(x^2 - y^2)$ gives real Gram matrices of rank three.

Remark 1.3. If A is real with nonnegative entries, every permutation term is nonnegative and (2) is immediate. A violation therefore requires the long-cycle contribution to be negative and large. The constructions arrange this through complex phases in rank two and through sign patterns in rank three.

2. The Fischer norm and Gram permanents

Let $\mathcal{F}_d$ be the Fock space of entire functions g on $\mathbb{C}^d$ with

$$\|g\|^2 = \pi^{-d} \int_{\mathbb{C}^d} |g(z)|^2 e^{-|z|^2}\, \mathrm{d}A(z) < \infty. \tag{6}$$

The monomials z^α are orthogonal in $\mathcal{F}_d$ with $\|z^\alpha\|^2 = \alpha! = \prod_k \alpha_k!$, and the inner product is invariant under unitary changes of the variables. For $v \in \mathbb{C}^d$ let $\ell_v(z) = \sum_k v_k z_k$, and let $\langle v, w\rangle = \sum_k \overline{v_k} w_k$.

Lemma 2.1. *For $v_1, \dots, v_m, w_1, \dots, w_m \in \mathbb{C}^d$,*

$$\left\langle \prod_{i=1}^m \ell_{v_i}, \prod_{j=1}^m \ell_{w_j} \right\rangle = \operatorname{per}\left[\langle v_i, w_j\rangle\right]_{i,j=1}^m. \tag{7}$$

Proof. Expanding, $\prod_i \ell_{v_i} = \sum_\varphi \prod_i v_{i,\varphi(i)} z_{\varphi(i)}$, the sum over all maps $\varphi : \{1,\dots,m\} \to \{1,\dots,d\}$, and similarly for the w_j with maps ψ. The monomials $\prod_i z_{\varphi(i)}$ and $\prod_j z_{\psi(j)}$ are orthogonal unless φ and ψ have the same multiset of values, of type α say, in which case their inner product is $\alpha!$. The maps ψ with the multiset of φ are exactly the $\varphi \circ \tau$, $\tau \in S_m$, and each arises from $\alpha!$ permutations τ. Hence

$$\left\langle \prod_i \ell_{v_i}, \prod_j \ell_{w_j} \right\rangle = \sum_\varphi \sum_{\tau \in S_m} \prod_i \overline{v_{i,\varphi(i)}} \prod_j w_{j,\varphi(\tau(j))} = \sum_{\tau \in S_m} \prod_i \left\langle v_i, w_{\tau^{-1}(i)} \right\rangle,$$

which is (7). □

Corollary 2.2. *Let $A = \left(\langle v_i, v_j\rangle\right)_{i,j=0}^{n-1}$ with $v_i \in \mathbb{C}^d$, let $\ell_i = \ell_{v_i}$ and $p = \prod_{i=1}^{n-1} \ell_i$. Then* $\operatorname{per} A = \|\ell_0 p\|^2$, $\operatorname{per} A\big(\hat{0}, \hat{j}\big) = \|p/\ell_j\|^2$, *and*

$$R(A) = \frac{\sum_{j=1}^{n-1} |\langle v_0, v_j\rangle|^2 \|p/\ell_j\|^2}{\|\ell_0 p\|^2}. \tag{8}$$

Every Hermitian positive semidefinite matrix is such a Gram matrix with $d = \operatorname{rank} A$, and a real symmetric one is the Gram matrix of vectors in $\mathbb{R}^d$.

3. A univariate ratio with logarithmic growth

For a nonzero polynomial $f \in \mathbb{C}[z]$ with roots $\alpha_1, \dots, \alpha_d$, listed with multiplicity, define $Q(f)$ by (5). It is unchanged when f is multiplied by a nonzero scalar. Let

$$f_L(z) = \prod_{k=1}^{L} \left(1 - \frac{z^4}{81k^2}\right), \qquad f_\infty(z) = \frac{\sin(\pi z^2/9)}{\pi z^2/9}, \tag{9}$$

so that $f_L \to f_\infty$ locally uniformly by the sine product $\sin(\pi u)/(\pi u) = \prod_{k\ge1}(1 - u^2/k^2)$ [4] with $u = z^2/9$. The roots of f_L are $\alpha_{k,j} = 3\sqrt{k}\, i^j$, $1 \le k \le L$, $0 \le j \le 3$, with $|\alpha_{k,j}|^2 = 9k$.

Lemma 3.1. *For every $L \le \infty$, every subproduct h of f_L (in particular f_L itself) satisfies $|h(z)| \le e^{\pi|z|^2/9}$ for all $z \in \mathbb{C}$.*

Proof. $\left|1 - z^4/81k^2\right| \le 1 + |z|^4/81k^2$, and $\prod_{k\ge1}(1 + t^2/k^2) = \sinh(\pi t)/(\pi t) \le e^{\pi t}$ with $t = |z|^2/9$. □

Since $2\pi/9 < 1$, Lemma 3.1 places f_∞ and $z f_\infty$ in $\mathcal{F}_1$, and the dominant $4|z|^2 e^{(2\pi/9-1)|z|^2}$ with dominated convergence gives

$$\|f_L - f_\infty\| \to 0, \quad \|z(f_L - f_\infty)\| \to 0 \quad (L \to \infty). \tag{10}$$

Both limiting norms are positive; from the Taylor series of f_∞,

$$\|f_\infty\|^2 = \sum_{k\ge 0} \left(\frac{\pi}{9}\right)^{4k} \frac{(4k)!}{(2k+1)!^2} = 1.0105\ldots,$$

$$\|zf_\infty\|^2 = \sum_{k\ge 0} \left(\frac{\pi}{9}\right)^{4k} \frac{(4k+1)!}{(2k+1)!^2} = 1.0560\ldots.$$

Proposition 3.2. *With $H_L = \sum_{k=1}^{L} 1/k$,*

$$\sum_{\alpha : f_L(\alpha)=0} \left\| \frac{f_L}{z-\alpha} \right\|^2 = \frac{4}{9} H_L \|f_L\|^2 + O(1), \tag{11}$$

the $O(1)$ being uniform in L. Consequently $Q(f_L) \sim c_0 \log L$ with $c_0 = (4/9)\|f_\infty\|^2/\|zf_\infty\|^2 = 0.4253\ldots$.

Proof. Fix a root $\alpha = \alpha_{k,j}$ and let $h = h_{L,k} = \prod_{l\ne k}(1 - z^4/81l^2)$, so that $f_L = (1 - z^4/\alpha^4)h$ because $\alpha^4 = 81k^2$. Since $1 - z^4/\alpha^4 = -(z-\alpha)(z^3 + \alpha z^2 + \alpha^2 z + \alpha^3)/\alpha^4$,

$$-\alpha \frac{f_L}{z-\alpha} = \left(1 + \frac{z}{\alpha} + \frac{z^2}{\alpha^2} + \frac{z^3}{\alpha^3}\right) h, \quad \text{hence} \quad -\alpha \frac{f_L}{z-\alpha} - f_L = \left(\sum_{r=1}^{4} \left(\frac{z}{\alpha}\right)^r\right) h.$$

By Lemma 3.1 the norms $\|z^r h\|$, $1 \le r \le 4$, are bounded by a constant independent of L and k, and $|\alpha| \ge 3$, so the right side has norm at most $C/|\alpha|$. Therefore $|\alpha|\|f_L/(z-\alpha)\| = \|f_L\| + O(|\alpha|^{-1})$, and since $\|f_L\|$ is bounded,

$$\left\| \frac{f_L}{z-\alpha} \right\|^2 = \frac{\|f_L\|^2}{|\alpha|^2} + O(|\alpha|^{-3}).$$

Summing over the four roots with $|\alpha|^2 = 9k$ and then over $k \le L$ gives $4\|f_L\|^2 \sum_k 1/9k$ plus an error bounded by a constant times $\sum_{k\ge 1} k^{-3/2} < \infty$. This is (11). Dividing by $\|zf_L\|^2$ and using (10) and $H_L \sim \log L$ gives the asymptotic for $Q(f_L)$. □

4. Complex matrices of rank two

Fix a polynomial $f = \sum_{k=0}^{d} c_k z^k$ of degree $d \ge 1$ with roots $\alpha_1, \ldots, \alpha_d$, and an integer $M \ge 1$. Let $A_{f,M}$ be the Gram matrix of the $1 + d + M$ vectors

$$v_0 = (1, 0), \quad v_j = \left(1, -\alpha_j/\sqrt{M}\right) \quad (1 \le j \le d), \quad v_{d+1} = \cdots = v_{d+M} = (0, 1) \tag{12}$$

in $\mathbb{C}^2$. It is Hermitian positive semidefinite of rank exactly two, with $\langle v_0, v_j \rangle = 1$ for $1 \le j \le d$ and $\langle v_0, v_i \rangle = 0$ for $i > d$. In the variables (z, w) the linear forms are $\ell_0 = z$, $\ell_j = z - \left(\alpha_j/\sqrt{M}\right)w$ and $\ell_i = w$ for $i > d$. Define the homogenization

$$H_{M,d} f(z, w) = \sum_{k=0}^{d} c_k M^{-(d-k)/2} z^k w^{M+d-k}. \tag{13}$$

Then $c_d \prod_{i\ge 1} \ell_i = H_{M,d} f$: indeed $w^M \prod_j \left(z - \alpha_j w/\sqrt{M}\right) = w^M \left(w/\sqrt{M}\right)^d f\left(z\sqrt{M}/w\right)/c_d$, and expanding f gives (13). Likewise, for $g_j = f/\left(z - \alpha_j\right)$, which has degree $d-1$ and leading coefficient c_d, $c_d \prod_{i\ge 1, i\ne j} \ell_i = H_{M,d-1} g_j$.

Lemma 4.1. *Put $b_{M,j} = (M+j)!/(M!\, M^j) = \prod_{h=1}^{j}(1 + h/M)$, so that $1 \le b_{M,j} \le \exp(j(j+1)/2M)$. For $g = \sum_k g_k z^k$ of degree m,*

$$\left\|zH_{M,m}g\right\|^2 = M!\sum_{k=0}^{m}(k+1)!|g_k|^2 b_{M,m-k}, \quad \left\|H_{M,m}g\right\|^2 = M!\sum_{k=0}^{m}k!|g_k|^2 b_{M,m-k}. \tag{14}$$

Proof. The monomials in $zH_{M,m}g$ are $z^{k+1}w^{M+m-k}$ with coefficients $g_k M^{-(m-k)/2}$; orthogonality gives $\sum_k |g_k|^2 M^{-(m-k)}(k+1)!(M+m-k)!$, which is the first formula. The second is the same computation without the factor z. □

Proposition 4.2.

$$e^{-d(d+1)/2M}Q(f) \le R\big(A_{f,M}\big) \le e^{d(d+1)/2M}Q(f). \tag{15}$$

Proof. By (8), the identification of the forms, and the vanishing of $\langle v_0, v_i\rangle$ for $i > d$,

$$R\big(A_{f,M}\big) = \frac{\sum_{j=1}^{d}\left\|H_{M,d-1}g_j\right\|^2}{\left\|zH_{M,d}f\right\|^2}.$$

By Lemma 4.1 the numerator lies between $M!\sum_j \left\|g_j\right\|^2$ and $e^{d(d-1)/2M}$ times that, and the denominator lies between $M!\|zf\|^2$ and $e^{d(d+1)/2M}$ times that. The quotient of these ranges is (15). □

Theorem 4.3. *Let $A_L = A_{f_L,(4L)^3}$, a complex Hermitian positive semidefinite matrix of rank two and order $n_L = 64L^3 + 4L + 1$. Then $R(A_L) \sim (c_0/3)\log n_L$ as $L \to \infty$. In particular R is unbounded over complex matrices of rank two.*

Proof. Here $d = 4L$ and $M = d^3$, so $d(d+1)/2M = (d+1)/2d^2 \to 0$ and Proposition 4.2 gives $R(A_L) \sim Q(f_L)$. Proposition 3.2 gives $Q(f_L) \sim c_0 \log L$, and $\log n_L = 3\log L + O(1)$. □

5. Real matrices of rank three

Fix f as before, with roots $\alpha_j = a_j + ib_j$, $a_j, b_j \in \mathbb{R}$, and an integer $M \ge 1$. Let $B_{f,M}$ be the Gram matrix of the $4M + d + 1$ real vectors

$$u_0 = (1,0,0), \quad u_j = \left(1, -\frac{a_j}{\sqrt{2M}}, -\frac{b_j}{\sqrt{2M}}\right) \quad (1 \le j \le d), \tag{16}$$

together with M copies of each of $(0,1,0)$, $(0,0,1)$, $(0,1,1)$, $(0,1,-1)$. It is real symmetric positive semidefinite of rank three, and $\langle u_0, u_j\rangle = 1$ for $1 \le j \le d$, $\langle u_0, u_i\rangle = 0$ for the remaining vectors. In the variables (z,x,y) the forms are $\ell_0 = z$, $\ell_j = z - \lambda_j$ with

$$\lambda_j = \frac{a_j x + b_j y}{\sqrt{2M}}, \tag{17}$$

and the product of the $4M$ remaining forms is $q(x,y)^M$ with $q(x,y) = xy(x^2 - y^2)$. Hence

$$R\big(B_{f,M}\big) = \frac{\sum_{j=1}^{d}\left\|q^M\prod_{l\ne j}(z-\lambda_l)\right\|^2}{\left\|z\,q^M\prod_{l=1}^{d}(z-\lambda_l)\right\|^2}. \tag{18}$$

Proposition 5.1. $\lim_{M\to\infty} R\big(B_{f,M}\big) = Q(f)$.

Proof. Write both norms in (18) as Gaussian integrals (6) over $(z,x,y) \in \mathbb{C}^3$. The factor $|q(x,y)|^{2M}e^{-|x|^2-|y|^2}$ is common to numerator and denominator; normalize it to a probability measure μ_M on $\mathbb{C}^2$, and let z be an independent standard complex Gaussian. Then

$$R(B_{f,M}) = \frac{\mathbb{E}\Big[\sum_j \prod_{l\neq j} |z-\lambda_l|^2\Big]}{\mathbb{E}\Big[|z|^2 \prod_l |z-\lambda_l|^2\Big]}, \tag{19}$$

the expectations taken over μ_M and z.

In polar coordinates $(x,y) = r\omega$ with ω on the unit sphere $S^3 \subset \mathbb{C}^2$, the measure μ_M has density proportional to $r^{8M+3}e^{-r^2}|q(\omega)|^{2M}$. So r^2 has the Gamma distribution with shape $4M+2$ and rate 1, independent of ω, and $\mathbb{E}\big[(r^2/4M)^s\big] \to 1$ for every fixed $s \geq 0$. The angular law ν_M has density proportional to $|q(\omega)|^{2M}$. On the sphere, with $a = |x|^2$, $b = |y|^2$, $a+b=1$ and $\theta = \arg x - \arg y$,

$$|q(x,y)|^2 = ab\,|x^2-y^2|^2 = ab(1-4ab\cos^2\theta) \leq \frac{1}{4},$$

with equality exactly on the two circles $C_\pm = \big\{e^{i\varphi}(1,\pm i)/\sqrt{2}\big\}$. If $K \subset S^3$ is closed and disjoint from $C_+ \cup C_-$, then $\max_K |q|^2 < 1/4 - \eta$ for some $\eta > 0$, while the set where $|q|^2 > 1/4 - \eta/2$ has positive surface measure; so $\nu_M(K) \to 0$. Thus every limit point of ν_M is supported on $C_+ \cup C_-$. The maps $\omega \mapsto e^{i\varphi}\omega$ and $(x,y) \mapsto (x,-y)$ preserve the surface measure and $|q|$, so they preserve every ν_M; the limit is therefore the measure that picks C_+ or C_- with probability $1/2$ each and a uniform phase φ on it. Since S^3 is compact and the test functions below are polynomials in $\omega, \overline{\omega}$, their ν_M-expectations converge, and together with the Gamma moments this gives convergence of every polynomial moment of $(x,y)/\sqrt{2M} = \big(r/\sqrt{2M}\big)\omega$ to the corresponding moment of $e^{i\varphi}(1,\varepsilon i)$, $\varepsilon = \pm 1$.

Under this limit $\lambda_j \to e^{i\varphi}\big(a_j + \varepsilon i b_j\big)$, which is $e^{i\varphi}\alpha_j$ for $\varepsilon = 1$ and $e^{i\varphi}\overline{\alpha_j}$ for $\varepsilon = -1$. The integrands in (19) are polynomials in $z, \overline{z}, \lambda, \overline{\lambda}$, so numerator and denominator converge to their values at the limit. Replacing z by $e^{i\varphi}z$, which leaves the Gaussian law of z unchanged, removes the phase, and replacing the roots by their conjugates conjugates the coefficients of the polynomials in z, which leaves Fischer norms unchanged. The denominator therefore tends to $\|zf\|^2/|c_d|^2$ and the numerator to $\sum_j \big\|f/(z-\alpha_j)\big\|^2/|c_d|^2$. The quotient is $Q(f)$. □

Theorem 5.2. *R is unbounded over real symmetric positive semidefinite matrices of rank three.*

Proof. Given K, Proposition 3.2 provides L with $Q(f_L) > K+1$, and Proposition 5.1 provides M with $R\big(B_{f_L,M}\big) > K$. □

6. Correlation matrices, integer entries, positive definiteness

Under a congruence $A \mapsto DAD^*$ with D invertible diagonal, $\operatorname{per} A$ and every term $\big|a_{0j}\big|^2 \operatorname{per} A\big(\hat{0},\hat{\jmath}\big)$ are multiplied by the same factor $\prod_i |d_{ii}|^2$, so R is unchanged. All matrices constructed above have positive diagonal entries, so they can be normalized to correlation matrices of the same rank and the same ratio. This proves the first half of Theorem 1.1(iii).

For integer entries in the real case, fix f_L and M with $R\big(B_{f_L,M}\big) > K$ and replace the vectors u_j of (16) by rational vectors, keeping u_0 and the padding vectors fixed. The ratio is a continuous function of the vectors, so a sufficiently fine approximation keeps $R > K$ and, because the fixed vectors span $\mathbb{R}^3$, keeps the rank equal to three. Scaling each vector by an integer is a diagonal congruence, so a common denominator produces an integer Gram matrix with the same ratio. This proves Theorem 1.1(ii).

Finally, if A is any fixed matrix with $R(A) > K$, then $A + \varepsilon I$ is positive definite and $R(A + \varepsilon I) > K$ for all sufficiently small $\varepsilon > 0$, by continuity; normalization to a correlation matrix preserves this. This completes Theorem 1.1(iii). The perturbation does not preserve low rank.

7. Explicit examples

Proof of Theorem 1.2. With the forms $\ell_0 = z$, $\ell_j = 5z + 3i^{j-1}w$ $(1 \le j \le 4)$, $\ell_{4+j} = 5z + 5(1+i)i^{j-1}w$ $(1 \le j \le 4)$ and $\ell_9 = \cdots = \ell_{12} = w$, the product of the twelve nondistinguished forms is

$$\begin{aligned} p(z,w) &= w^4(625z^4 - 81w^4)(625z^4 + 2500w^4) \\ &= 390625\, z^8 w^4 + 1511875\, z^4 w^8 - 202500\, w^{12}, \end{aligned}$$

since $\prod_j (5z + 3i^j w) = (5z)^4 - (3w)^4$ and $\prod_j (5z + 5(1+i)i^j w) = (5z)^4 - (5(1+i))^4 w^4$ with $(1+i)^4 = -4$. By Corollary 2.2, $\operatorname{per} A = \|zp\|^2 = 390625^2 \cdot 9! \cdot 4! + 1511875^2 \cdot 5! \cdot 8! + 202500^2 \cdot 12!$, which is the value in (4). The Gram entries are $\langle v_0, v_j \rangle = 5$ for $1 \le j \le 8$ and 0 for the four copies of $(0,1)$, so $T(A) = 25 \sum_{j=1}^{8} \|p/\ell_j\|^2$. Multiplication of w by i is a unitary change of variables that permutes the forms within each quartet, so the eight norms take only two values, one for each quartet:

$$N_1 = \|p/\ell_1\|^2 = 277\,968\,834\,000\,000\,000, \quad N_2 = \|p/\ell_5\|^2 = 44\,206\,998\,210\,000\,000,$$
$$T(A) = 100(N_1 + N_2),$$

and expanding $p/\ell_1 = w^4(5z + 3iw)(5z - 3w)(5z - 3iw)(625z^4 + 2500w^4)$ and p/ℓ_5 into monomials gives the value of $T(A)$ in (4). The difference is $T(A) - \operatorname{per} A = 187\,167\,321\,000\,000\,000$. The rank is two because the vectors span $\mathbb{C}^2$ and lie in it. $\square$

The same matrix satisfies $(\operatorname{per} A)^2 - \operatorname{per}\big(A \circ \overline{A}\big) > 0$: it is not a counterexample to Chollet's conjecture [5], and the two-cycle inequality (2) is genuinely weaker than the inequalities it was derived from.

Theorem 7.1. *Let B be the Gram matrix of the real vectors $(1,0,0)$, distinguished; $(8,\pm 1,0)$ and $(8,0,\pm 1)$; and 117 copies of each of $(0,1,0)$, $(0,0,1)$, $(0,1,1)$, $(0,1,-1)$. Then B is a real symmetric positive semidefinite integer matrix of order 473 and rank three with*

$$R(B) = 1.002580227152924020\ldots > 1, \quad \text{in particular} \quad \frac{401}{400} < R(B) < \frac{5013}{5000}. \tag{20}$$

Proof. The product of the nondistinguished forms is $p = q(x,y)^{117}(64z^2 - x^2)(64z^2 - y^2)$ with $q = xy(x^2 - y^2)$, whose expansion $x^{117}y^{117}(x^2-y^2)^{117}(4096z^4 - 64z^2(x^2+y^2) + x^2y^2)$ has at most 354 monomials. By Corollary 2.2, $\operatorname{per} B = \|zp\|^2$ and $T(B) = 64 \sum_{j=1}^{4} \|p/\ell_j\|^2$, where ℓ_j runs over $8z \pm x$, $8z \pm y$; each norm is a finite sum of squared integer coefficients times factorials, and exact rational arithmetic gives (20). The permanent $\operatorname{per} B$ has 988 decimal digits. $\square$

8. Computational confirmation

The values $Q(f_L)$ computed from (5) are $0.4993, 0.7249, 0.9792, 1.2507, 1.5321, 1.8194$ for $L = 1, 2, 4, 8, 16, 32$; the increments $Q(f_{2L}) - Q(f_L)$ are $0.2543, 0.2715, 0.2815, 0.2873$, approaching $c_0 \log 2 = 0.2948$. For $f = f_1$ the ratios $R\big(B_{f_1, M}\big)$ are $0.2483, 0.3316, 0.3974, 0.4422, 0.4690, 0.4837$ for $M = 1, 2, 4, 8, 16, 32$, increasing to $Q(f_1) = 0.4993$. A randomized adversarial search for large R over Hermitian positive semidefinite matrices of orders $3 \le n \le 8$, all ranks, real and complex, found no violation of (2); the largest value found

was $R = 0.793$, for a complex matrix of rank two and order 8, and at order 4 nothing above 0.543. The values in Theorems 1.2 and 7.1 were computed twice, by Ryser's formula over the Gaussian integers on the full matrix and its principal minors, and by the sparse Fischer-norm expansions in Section 7; the two computations agree exactly. None of these computations enters the proofs.

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